

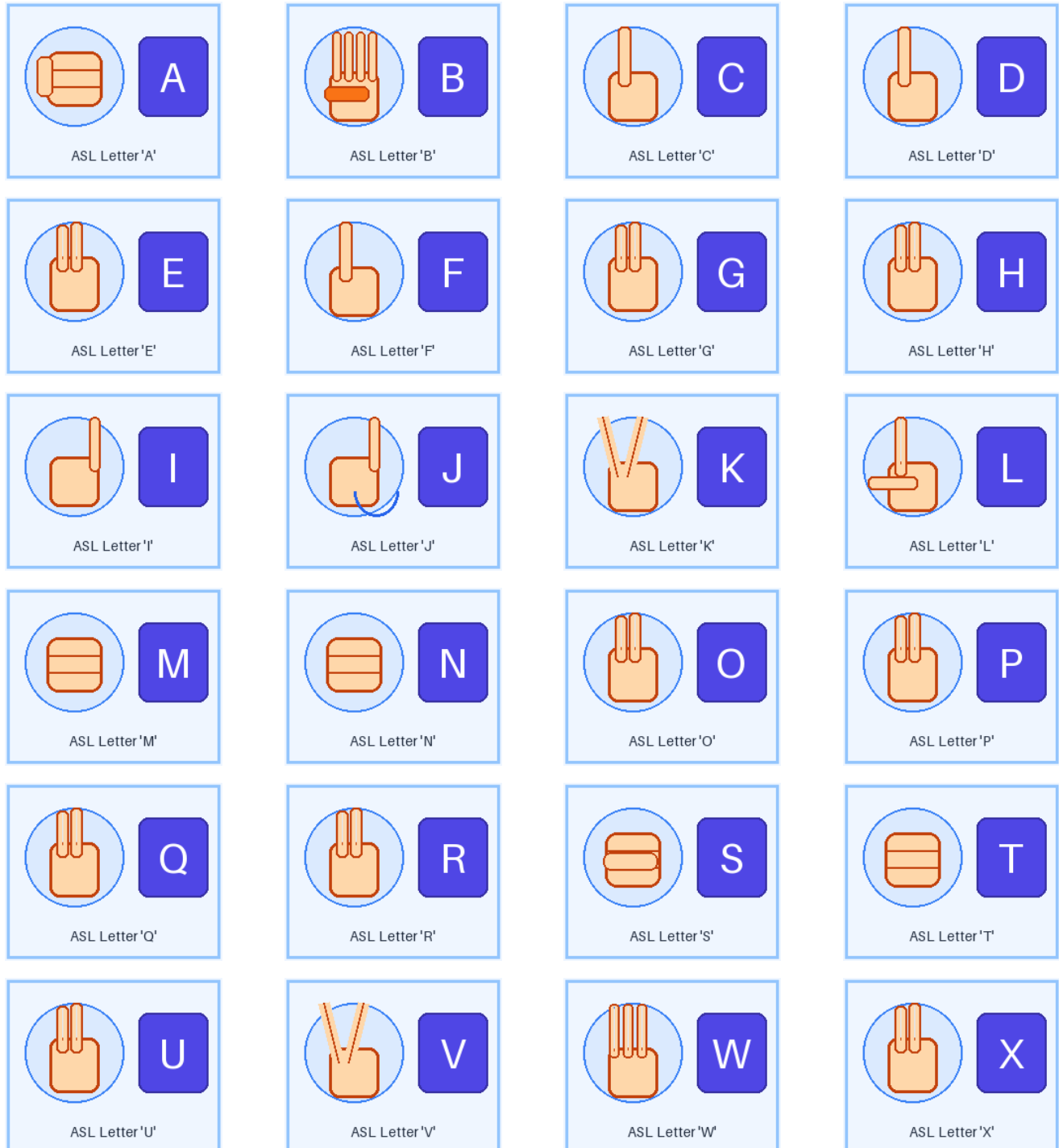
■ SignBridge AI

Official Sign Language Illustrated Handbook & Visual Lexicon

Authoritative Visual Reference Manual with Anatomical Formations, Motion Vectors & AI Protocol

■ Section 1: Standard Fingerspelling Alphabet (A–Z Visual Gallery)

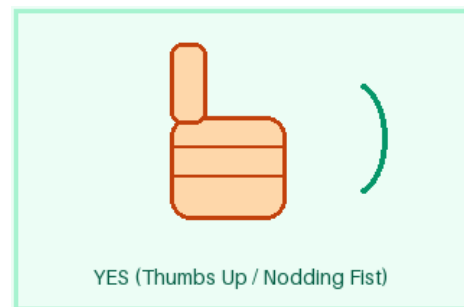
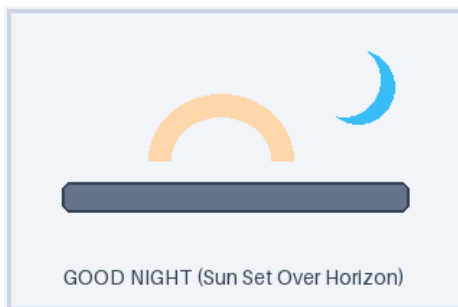
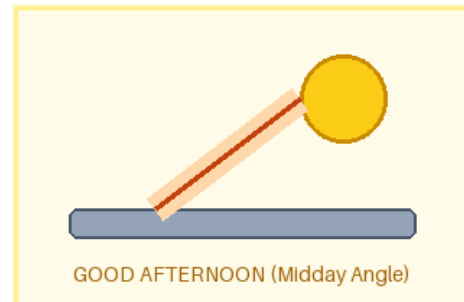
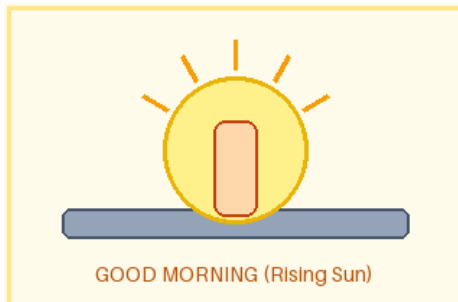
Standardized manual alphabet formations for spelling proper nouns, technical terms, and fingerspelling.





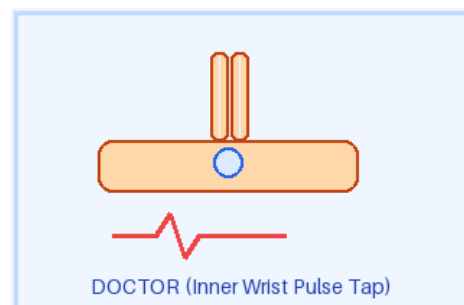
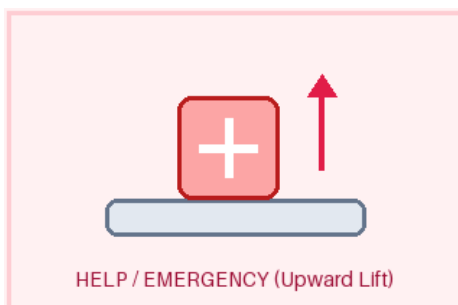
■ Section 2: Core Greetings & Civility Signs (Visual Guide)

Essential daily conversational signs with exact motion trajectories and hand postures.



■ Section 3: Emergency, Healthcare & Expressive Signs

Critical emergency triage and expressive signs recognized by the SignBridge system.





I LOVE YOU (Thumb + Index + Pinky)



GOOD MORNING (Rising Sun)

■■ Section 4: Vision AI Architecture & 2.0-Second Hold Gate Protocol



- 1. 2.0-Second Temporal Hysteresis Gate:** To eliminate false positives during hand motion and transitions, the SignBridge engine tracks continuous pose stability. A sign must be maintained steadily for 2.0 full seconds (2000 ms) with high confidence before it commits into the translated sentence and vocalizes speech.
- 2. 21 3D Landmark Tracking:** Runs client-side MediaPipe Hands at 60 FPS without server latency, computing 24-D spatial feature vectors across finger extensions, pinches, spreads, and elevations.
- 3. Automated Speech Synthesis:** When a sign reaches the 2.0-second commit gate, the Web Speech API automatically pronounces the translated word aloud and appends it with proper grammatical capitalization.